

Week 3 — MMM Field Guide

The equation • the transforms • the controls a pharma MMM needs • a fitting checklist • ROI definitions • the memo template • vocabulary

Applied Data Science and AI in Pharma Commercial • Columbia University • Fall 2026

Student reference — keep it next to Homework 1 and every MMM you build this semester

Companion to the Week 1 glossary and the Week 2 Claims Data Field Guide

How to use this guide. Part 1 is the model, term by term. Part 2 is what goes into a pharma MMM — channels, doses and the controls that consumer-goods templates leave out. Part 3 is the checklist you run before you show anyone a coefficient. Part 4 defines the ROI numbers and the reallocation rule. Part 5 is the memo template the homework asks for. Part 6 is vocabulary. Vendor and library names are examples, not endorsements.

Part 1 • The model, term by term

$TRx_t = \beta_0 + \sum_c \beta_c \cdot \text{Hill}(\text{Adstock}(x_{c,t}; \delta_c); K_c, s_c) + \gamma \cdot \text{ramp}_t + \eta_1 \cdot \sin_t + \eta_2 \cdot \cos_t + \lambda \cdot \text{competitor}_t + \phi \cdot \text{access}_t + \epsilon_t$

Term	What it means	How it is estimated	Typical values / notes
$\beta_0, \gamma \cdot \text{ramp}$	The base: what the brand does with zero promotion this week — awareness, refills accumulating, access	OLS; ramp midpoint and scale on a small grid (AIC)	Base share 50–80% of TRx for an in-line brand; higher for mature brands
$\text{Adstock}(x; \delta)$	Carryover: $a_t = x_t + \delta \cdot a_{t-1}$. δ = share of last week's effect that survives	Grid or prior (non-linear)	TV 0.6–0.8 • detailing 0.3–0.6 • conferences 0.3–0.5 • email/search 0–0.25. Half-life = $\ln 0.5 / \ln \delta$
Weibull adstock	Carryover whose peak can come after week 0 (action lag)	Grid or prior	Use when the detail → script → fill → claim lag is weeks (chronic Rx)
$\text{Hill}(a; K, s)$	Saturation: $a^s / (a^s + K^s) \in [0, 1]$. K = half-saturation point, s = shape	Grid or prior (non-linear)	K as a multiple of median adstocked spend (0.5–8×); $s \in [1, 3]$; $s > 1$ = S-curve with a threshold
Logistic saturation	$1 - \exp(-\lambda \cdot a)$: concave from zero, one parameter	Grid or prior	PyMC-Marketing's default; no threshold
β_c	Maximum weekly TRx the channel can deliver (Hill tops out at 1)	OLS / posterior	Must be positive; a negative β is a confounder, not a finding
season	Annual sine + cosine (or month dummies)	OLS	Deductible resets, refill timing, flu season; enter before channels
competitor	Step or ramped indicator from the launch week; or the competitor's own promotion	OLS	A –5 to –20% share effect is common in the first year
access	Share of lives with preferred status; PA/step-edit flags; tier changes	OLS	Comes from the payer-intelligence file (Week 2, category 4), not from the agency
ϵ	Everything not modeled: copay program, samples, price, MLR timing, panel restatements	Residual diagnostics	Structure in ϵ means something is missing; inspect before adding flexibility

Part 2 • What goes into a pharma MMM

2.1 Channels and their 'dose'

A dollar is not the same dose in every channel. Choose the input that best measures exposure; keep spend for the ROI denominator.

Channel	Dose to model	Spend pattern	Identification	Notes
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HCP detailing (personal promotion)	Calls or call-equivalents (detail = 1, sample drop = 0.3, lunch-and-learn = 2) or reach × frequency	Steady; holiday dips; step changes when the field is resized	Weak — steady spend gives the curve nothing to learn from	The largest budget and the least measurable; territory holdouts are the fix
Samples	Units dropped	Tracks detailing	Confounded with detailing	Separate lever; has gross-to-net implications (Week 1)
Conferences / congresses	\$ or attendee-reach	Spikes (4–6 a year)	Good for β , weak for δ	Model by event; compute ROI per event, not per week
Non-personal promotion (NPP): email, banners, journal ads	Sends / impressions delivered	Waves; approved-content calendars	Good if there are dark weeks	Approved (MLR) content limits how fast you can scale
Paid search, social, programmatic	Impressions or clicks	Pulses; co-spend with TV	Good with pulses; watch VIF with TV	HIPAA-compliant tracking only; no patient-level attribution
DTC TV / CTV	GRPs or impressions	Flights of 6–10 weeks	Excellent — on/off is the ideal pattern	Long δ ; MLR review 6–10 weeks; geo tests are standard
Speaker programs, KOL, peer-to-peer	Attendees	Episodic	Poor at national level	Better measured at HCP level (Weeks 4–5)
Copay program	Enrolled patients / redemptions	Continuous	Confounded with access	Not media — a separate lever with net-price cost

2.2 Controls that a consumer-goods template leaves out

- **Access:** formulary tier, prior-auth and step-edit flags, share of covered lives with preferred status, Medicare Part D vs. commercial mix. A tier change moves TRx overnight with zero spend change.
- **Competitor events:** launches, LOE/generic entry, label changes, recalls, DTC bursts by competitors.
- **Own events:** label expansions, indication launches, price changes, supply issues, field-force resizing.
- **Data artefacts:** panel restatements (Week 2), holiday weeks, 53-week years, projection methodology changes — dummy them or the model will credit a channel.
- **Seasonality:** annual sine/cosine or month dummies; entered before any channel.

2.3 The response variable

Weekly projected TRx (or NBRx for a launch brand, or units for an infused product) at national or geo level. Know how it was projected (Week 2): a panel change mid-series will look like a marketing effect. For geo-level MMM, use DMA or state × week; the extra rows help identification enormously.

Part 3 • The ten-point checklist before you show a coefficient

1. **Plot everything first.** TRx, each channel's spend, same-week scatter. Mark events. If you cannot see the launch ramp and the seasonality by eye, the model will not either.
2. **Controls in before channels.** Ramp, season, competitor, access. Fit the ramp media-blind, then refine on AIC.
3. **Bound the grid with knowledge.** Per-channel δ ranges; K relative to the channel's own spend; s between 1 and 3. Record the grid in the appendix.
4. **Search one channel at a time, sweep twice.** Coordinate descent on AIC (or BIC for a smaller model). Keep the top three cells per channel.
5. **Signs and magnitudes.** Every media $\beta > 0$. β_c is the maximum weekly TRx the channel can deliver — sanity-check it against the brand's size.
6. **Residuals.** Plot over time and by month; Durbin–Watson near 2; no slow waves; no spikes at known events you forgot.

- 7. VIF.** < 5 fine, 5–10 suspicious, > 10 not separately identified. Co-spend and ramp-like channels are the usual causes; fix by bounding δ , merging channels, ridge, or an experiment.
- 8. Hold-out.** Train on the first two-thirds, predict the last third; report MAPE and say whether the grid was re-run on the training window. State what the model cannot forecast (events it never saw).
- 9. Ranges.** Block-bootstrap the residuals (8–13-week blocks) for coefficient uncertainty; re-run the runner-up grid cells for specification uncertainty. Report both on one chart.
- 10. Calibrate.** Compare at least one channel with an experiment (geo holdout, territory test, conversion lift). Where they disagree, the experiment wins.

Part 4 • ROI, marginal ROI and the reallocation rule

Quantity	Definition	Use	Trap
Incremental TRx (channel)	$\sum_t \beta_c \cdot \text{Hill}(\text{Adstock}(x_{c,t}))$	Decomposition; share of TRx from promotion	Depends on the base being right
Average ROI	$(\text{incremental TRx} \times \text{contribution per TRx}) \div \text{spend}$	Reporting: 'what did we get for what we spent'	A saturated channel can have the best average and the worst marginal
Cost per incremental TRx	$\text{spend} \div \text{incremental TRx}$	The same thing inverted; easier for non-quant audiences	—
Marginal ROI	$\Delta(\text{contribution}) \div \Delta(\text{spend})$ for a small change (e.g. +5% in every week)	Decisions: 'where does the next dollar go'	A slope at a point; says nothing about +400%
Contribution per TRx	Net revenue per TRx minus COGS and variable cost (Week 1 P&L)	The ROI numerator	Finance owns it; never invent it; it differs by channel mix (payer)
Payback / break-even	$\text{Spend} \div (\text{incremental TRx} \times \text{contribution})$ with carryover counted	Long-tail channels (TV) need the tail counted	Truncating adstock understates slow channels

The reallocation rule

At the optimum, every channel has the same marginal ROI. Move money from the lowest to the highest marginal channel in small steps (e.g. \$0.25M), re-evaluate the curves after each step, and stop when the gap is smaller than your uncertainty. Always with constraints:

- **Floors:** field-force size, contracted sponsorships, minimum presence in a launch year.
- **Caps:** inventory (TV slots), list size and fatigue (email), MLR-approved content (digital), hiring (detailing).
- **Confidence:** do not move money on a 0.1 gap when the interval is ± 0.5 .
- **Strategy and compliance:** launch awareness goals, DTC fair-balance requirements, review cycles.

The homework question in one line

'Which \$10M would you cut?' is a marginal question with a fixed total. Answer it with two scenarios (conservative / aggressive), the expected TRx change with a range, and the constraint that binds first.

Part 5 • The memo to the Brand Lead — template (2 pages)

Section	Length	What goes in it
TL;DR	1 paragraph	The recommendation in numbers: what to cut, what to grow, expected TRx effect with a range, and the one thing you are least sure of
What drove TRx	1 figure + 3 lines	The decomposition chart; base vs. incremental share; the two events
ROI by channel	1 small table	Average and marginal, with ranges; rank; one sentence per channel
The reallocation	1 table	Current → proposed by channel for two risk postures; the constraint that binds; feasibility (MLR, contracts, hiring)
What the model does not capture	4–6 bullets	Copay program, samples, price, MLR timing, panel restatements, anything with no variation in the data

The experiment	3 lines	Design (units, duration, dark share), what it resolves, cost in spend and lost TRx
Appendix (notebook)	—	Grid, AIC table, diagnostics, hold-out, bootstrap; AI-use paragraph (syllabus)

Part 6 • Vocabulary

Term	Meaning
Adstock / carryover	The lagged effect of spend; geometric (one decay parameter) or Weibull (delayed peak)
Saturation / diminishing returns	Response per dollar falls as spend rises; Hill or logistic transforms
Response curve	Contribution as a function of (adstocked) spend for one channel; adstock decides when, saturation decides how much
Base vs. incremental	What the brand does with zero promotion vs. what promotion added; base includes access and competitor effects
Decomposition / contribution	Weekly TRx split into base and per-channel bands; the chart leadership wants
Average vs. marginal ROI	Total return ÷ total spend vs. the return on the next dollar; saturation makes them rank channels differently
Identifiability	Whether the data can distinguish parameter combinations; δ and β trade off; two fits with equal AIC can imply different ROI
VIF	Variance inflation factor — how much a regressor's variance is inflated by collinearity with the others
Hold-out / back-test	Fit on early weeks, predict later weeks; tests the coefficients (and, if re-run, the grid)
Block bootstrap	Resample residuals in contiguous blocks to keep autocorrelation, refit, read percentiles
Geo-lift / holdout	An experiment where a channel is dark in some geographies; difference-in-differences gives causal lift
Difference-in-differences	(exposed test – exposed pre) – (control test – control pre), or the growth-ratio version
Calibration	Constraining or checking an MMM against experimental lift; as a prior in a Bayesian MMM
Prior / posterior / credible interval	Bayesian terms: belief before data, belief after, and the interval containing X% of the posterior
MTA	Multi-touch attribution: person-level credit for digital touchpoints; blind offline; privacy-limited in pharma
Interrupted time series	Model for a one-off event (label change, recall): level and slope change at a known date
GRP / impressions / reach / frequency	TV and digital dose measures: gross rating points; ad exposures; unique audience; exposures per person
NPP	Non-personal promotion: email, banners, journal ads, web — everything that is not a rep
MLR	Medical-legal-regulatory review of promotional content; 6–10 weeks for DTC; a constraint on reallocation speed
Contribution per TRx	Net revenue per TRx less variable cost; the ROI numerator; from finance (Week 1 P&L)
QBR	Quarterly business review — where the Brand Lead's question is asked